Activity No: 4.3	
Activity Title: Sorting and Searching Arrays	
Course Code: CPE 007	**Program:** Computer Engineering
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Objectives

This activity aims to develop programming skills using arrays.

Intended Learning Outcomes (ILOs):
The students should be able to:

2.1 Analyse and create programs that sorts arrays

2.2 Analyse and create programs that search arrays

Instructions

Write a program that asks for a number from the user and prints which day of the week that number corresponds to. The days are indexed from 0 (Sunday) to 6 (Saturday). Before the program gets a value from the array, it must first check if the given day is greater than or equal to zero and less than 7. If not, it should print the message: "Error, no such day." Your version of the program must print the same result as the expected output.
Sample Input and Output:

Example input 0

Example output: Sunday

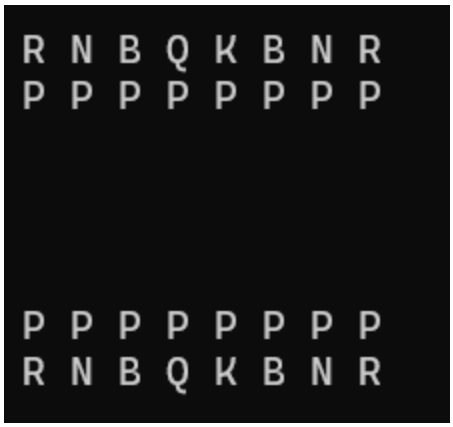
Example input 5

Example output: Friday

Example input 12

Example output: Error, no such day.

Write a program that creates a chessboard, sets all the pieces on it and then displays the contents of the board. Create a two-dimensional array, fill it with data and print a letter when a piece is on the field and a space when there is no piece. Store one letter for one piece. For now, we don't need any information about the color of the pieces. The starting positions (with letters which symbolize each piece) for all pieces are: The rooks (R) are placed on the outside corners, right and left edge (white on the 1st and black on the 8th line). The knights (N) are placed immediately inside of the rooks. The bishops (B) are placed immediately inside of the knights. The queen (Q) is placed on the central square of the same color as that of the player: white queen on the white square and black queen on the black square. Both stand on the d rank: white queen on the d1 field and black queen on the d8 field. The king (K) takes the vacant spot next to the queen. The pawns (P - not the official symbol, but you need to print something) are placed one square in front of all of the other pieces. Your version of the program must print the same result as the expected output.



Output/Answer

1.

```cpp
hoa4.3.1.cpp > main()
#include <iostream>
#include <cstring>
#include <string>
using namespace std;

int main () {

    string days[7] = {  "Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"};
    int num;

    cout << "Input a number from 0 - 6: ";
    cin >> num;

    if (num >= 0 && num < 7) {
        cout << "Your day is: " << days[num];

    } else {
        cout << "Error, no such day";
    }

    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

Error, no such day
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\hoa4.3.1.exe
Input a number from 0 - 6: 0
Your day is: Sunday
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\hoa4.3.1.exe
Input a number from 0 - 6: 1
Your day is: Monday
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\hoa4.3.1.exe
Input a number from 0 - 6: 6
Your day is: Saturday
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\hoa4.3.1.exe
Input a number from 0 - 6: 12
Error, no such day
```

EXPLANATION: This program uses a string array named `days` that stores the seven days of the week from Sunday to Saturday. It then asks the user to enter a number and stores it in the integer variable `num`. The `if` statement checks if the number is between 0 and 6, so the program can safely access the corresponding element in the array using `days[num]`. If the number is within the valid range, it displays the day that corresponds to that number. If the number is not between 0 and 6, it displays `"Error, no such day"` instead.

2.

```cpp
#include <iostream>
#include <cstring>
#include <string>
using namespace std;

int main () {

    char chess[8][8] = {
    {'R', 'N', 'B', 'Q', 'K', 'B', 'N', 'R'},
    {'P', 'P', 'P', 'P', 'P', 'P', 'P', 'P'},
    {' ', ' ', ' ', ' ', ' ', ' ', ' ', ' '},
    {' ', ' ', ' ', ' ', ' ', ' ', ' ', ' '},
    {' ', ' ', ' ', ' ', ' ', ' ', ' ', ' '},
    {' ', ' ', ' ', ' ', ' ', ' ', ' ', ' '},
    {'P', 'P', 'P', 'P', 'P', 'P', 'P', 'P'},
    {'R', 'N', 'B', 'Q', 'K', 'B', 'N', 'R'}

    };

    for (int g = 0; g < 8; g++) {
        for (int w = 0; w < 8; w++) {
            cout << chess[g][w] << " ";
        }

        cout << endl;
    }

    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\hoa4.3.exe
R N B Q K B N R
P P P P P P P P




P P P P P P P P
R N B Q K B N R
PS C:\CPE11S4-DOMINGO>
```

EXPLANATION: This program uses an 8 by 8 two-dimensional character array called `chess` to create a chessboard. Each letter represents a chess piece, such as `R` for rook, `N` for knight, `B` for bishop, `Q` for queen, `K` for king, and `P` for pawn, while the empty spaces are represented by `' '`. The pieces are already placed in their starting positions inside the array. The nested `for` loops are used to go through each row and column of the board and access each element using `chess[g][w]`. After checking each element, the program prints it with a space, and `endl` is used to move to the next row.

Supplementary Activity (if any)

Conclusion

In this activity, I learned how to use a two-dimensional array to store and organize data in rows and columns. I also learned how nested `for` loops can be used to access each element of the array. It helped me understand how arrays can be used to represent something like a chessboard and how each index corresponds to a specific position. Overall, I became more familiar with using 2D arrays and nested loops in C++.

Assessment Rubric: SO 7